

Hector Martinez

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EDUCATION

Clemson University

Clemson, SC

Bachelor of Science in Mathematical Sciences (Computer Science Emphasis)

Expected May 2028

- **Current Coursework (Fall 2026):** Intro to Programming (Python), Intro to Linear Algebra (MATH 3110), Intro to Mathematical Proofs (MATH 3190).
- Rigorous, proof-based mathematics foundation with a growing interest in quantitative applications, including actuarial science and financial mathematics.

Greenville Technical College

Greenville, SC

Undergraduate Transfer Coursework in Mathematics & Engineering

May 2024 – Aug 2026

- **Completed STEM Coursework:** Calculus I–III, Differential Equations, Physics I & II (Calculus-based Mechanics & Electromagnetism), Principles of Microeconomics, Engineering Graphics (SolidWorks).

TECHNICAL & ANALYTICAL SKILLS

Analytical Focus: Differential Equations, Linear Systems, Discrete Proof Formulation, Numerical Methods, Vector Mathematics

Software & Tools: VS Code, Git, GitHub, LaTeX / Overleaf, Microsoft Excel, SolidWorks

Programming Languages: Lua / Luau (multi-year experience), Python (in progress)

EXPOSITORY MATHEMATICS & TECHNICAL PROJECTS

Expository Mathematics & Problem Analysis

May 2026 – Present

github.com/MartinezMathematics/MathPapers

- Derived an exact closed-form product solution to the Putnam 2014 A2 determinant problem using discrete polynomial differences and recursive ratio decomposition.
- Authored formal manuscripts including “*An Approach to Non-Linear Differential Equations*”, linearizing a Bernoulli equation via coordinate mapping, Chain Rule dimension-lifting, and Variation of Parameters.
- Derived Euler’s Method, RK4, and the linear integrating-factor technique from first principles, and formulated proofs by contradiction for function-space linear independence.

Interactive Systems Engineering

2023 – Present

Luau / Roblox Platform, Client-Server Architecture, Vector Math

- Engineered event-driven state machines and client-server replication logic (RemoteEvents), applying spatial 3D vector math and raycasting.
- Profiled runtime memory usage and script execution cycles to maintain stable performance and eliminate network desynchronization.

PROFESSIONAL EXPERIENCE

Publix Super Markets

Clemson, SC

Deli Associate / Customer Service

Aug 2026 – Present

- Execute high-volume food preparation and custom orders during peak commercial rush periods under strict sanitation standards.
- Deliver fast, professional customer assistance, coordinating with kitchen staff to manage prep queues and eliminate bottlenecks.
- Maintain strict punctuality and consistent weekly shifts while managing a full-time mathematical sciences course load.

Dollar General

Greenville, SC

Lead Sales Associate

Mar 2023 – Aug 2026

- Demonstrated sustained operational reliability and leadership across a continuous 3+ year tenure.
- Managed floor inventory logistics, shelf merchandising, stockroom organization, and point-of-sale register reconciliation.
- Supervised store operations during high-volume and solo shifts, resolving customer escalations and training incoming associates.